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SEMICONDUCTOR DEVICE AND METHOD OF FABRICATING A SEMICONDUCTOR DEVICE

Abstract

A semiconductor device and a method of fabricating the same are provided. The method includes steps of forming a source region and a drain region in a substrate; forming a gate structure on the substrate, wherein the source region and the drain region are disposed on opposite sides of the gate structure and separated from the gate structure by a distance; depositing an inter-layer dielectric (ILD) layer over the substrate and the gate structure; forming a first trench in the ILD layer, wherein the first trench exposes a first portion of the gate structure and overlies an area between the gate structure and one of the source and drain regions from a top-view perspective; and depositing a first conductive material in the first trench to form a conductive plate, wherein the conductive plate is electrically connected to the gate structure.

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Background/Summary

BACKGROUND

[0001] As semiconductor technologies evolve, metal oxide semiconductor (MOS) transistors are increasingly used in today's integrated circuits. The MOS transistors are voltage controlled devices. When a control voltage is applied to a gate of a MOS transistor, and the control voltage is greater than a threshold of the MOS transistor, a conductive channel is established between a drain and a source of the MOS transistor. As a result, a current flows between the drain and the source of the MOS transistor. In contrast, when the control voltage applied to the gate of the MOS transistor is less than the threshold of the MOS transistor, the MOS transistor is turned off accordingly.

[0002] MOS transistors are increasingly important due to growing demand for battery-operated portable devices, such as cell phones. Maximizing battery life requires minimizing current flow, ideally with no current flow, during an off state of the MOS transistor so as to minimize or eliminate unnecessary power consumption.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 a schematic top view of a semiconductor device, in accordance with some embodiments of the present disclosure.

[0005] FIG. 2 is a schematic cross-sectional view along a line A-A' of the semiconductor device in FIG. 1.

[0006] FIG. 3 is a schematic cross-sectional view along a line B-B' of the semiconductor device in FIG. 1.

[0007] FIG. 4 a schematic top view of a semiconductor device, in accordance with some embodiments of the present disclosure.

[0008] FIG. 5 is a schematic cross-sectional view along a line C-C' of the semiconductor device in FIG. 4.

[0009] FIG. 6 is a schematic cross-sectional view along a line D-D' of the semiconductor device in FIG. 4.

[0010] FIG. 7 is a flowchart of a method of manufacturing a semiconductor device in accordance with some embodiments of the present disclosure.

[0011] FIGS. 8 to 12 are cross-sectional views of intermediate stages of the method of manufacturing a semiconductor device, in accordance with some embodiments of the present disclosure.

[0012] FIG. 13 is a top view of an intermediate stage of the method of manufacturing a semiconductor device, in accordance with some embodiments of the present disclosure.

[0013] FIG. 14 is a schematic cross-sectional view along a line E-E' of the semiconductor device in

FIG. 13.

[0014] FIGS. 15 to 20 are cross-sectional views of intermediate stages of the method of manufacturing a semiconductor device, in accordance with some embodiments of the present disclosure.

[0015] FIG. 21 is a top view of an intermediate stage of the method of manufacturing a semiconductor device, in accordance with some embodiments of the present disclosure.

[0016] FIG. 22 is a schematic cross-sectional view along a line F-F' of the semiconductor device in FIG. 21.

[0017] FIG. 23 is a schematic cross-sectional view along a line G-G' of the semiconductor device in FIG. 21.

[0018] FIG. 24 is a top view of an intermediate stage of the method of manufacturing a semiconductor device, in accordance with some embodiments of the present disclosure.

[0019] FIG. 25 is a schematic cross-sectional view along a line H-H' of the semiconductor device in FIG. 24.

[0020] FIG. 26 is a schematic cross-sectional view along a line I-I' of the semiconductor device in FIG. 24.

[0021] FIG. 27 is a top view of an intermediate stage of the method of manufacturing a semiconductor device, in accordance with some embodiments of the present disclosure.

[0022] FIG. 28 is a schematic cross-sectional view along a line J-J' of the semiconductor device in FIG. 27.

[0023] FIGS. 29 to 33 are cross-sectional views of intermediate stages of the method of manufacturing a semiconductor device in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0024] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0025] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0026] As used herein, the terms such as “first,” “second” and “third” describe various elements, components, regions, layers and/or sections, but these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another. The terms such as “first,” “second” and “third” when used herein do not imply a sequence, order, or importance unless clearly indicated by the context.

[0027] Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are

reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the normal deviation found in the respective testing measurements. Also, as used herein, the terms “substantially,” “approximately” or “about” generally mean within a value or range (e.g., within 10%, 5%, 1%, or 0.5% of a given value or range) that can be contemplated by people having ordinary skill in the art. Alternatively, the terms “substantially,” “approximately” or “about” mean within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of time, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by the terms “substantially,” “approximately” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another end point or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

[0028] In accordance with some embodiments of the present disclosure, FIG. 1 is a schematic top view of a semiconductor device **10** in accordance with some embodiments of the present disclosure, FIG. 2 is a schematic cross-sectional view along a line A-A' of the semiconductor device **10** in FIG. 1, and FIG. 3 is a schematic cross-sectional view along a line B-B' of the semiconductor device **10** in FIG. 1. Referring to FIGS. 1 to 3, the semiconductor device **10** includes a substrate **100**, a gate dielectric **232**, a gate electrode **246**, a source region **260A**, a drain region **260B**, a first conductive contact **290**, a plurality of second conductive contacts **294**, a plurality of conductive plates **292**, a first interconnect structure **340**, and a second interconnect structure **342**.

[0029] The gate dielectric **232** and the gate electrode **246** are stacked on an upper surface **102** of the substrate **100**. The gate dielectric **232** is sandwiched between the substrate **100** and the gate electrode **246**. The gate electrode **246** may be a metal gate, although the gate electrode **246** may include polysilicon, metal silicide, or the like. The source region **260A** and the drain region **260B** are disposed in the substrate **100** and on opposite sides of the gate electrode **246**. In some embodiments, the source region **260A** is separated from the gate electrode **246** by a distance **S** from a top-view perspective. Similarly, the drain region **260B** is separated from the gate electrode **246** by the distance **S** from the top-view perspective.

[0030] The first conductive contact **290** is disposed on the gate electrode **246**. In some embodiments, the first conductive contact **290** is physically and electrically connected to the gate electrode **246**. The first interconnect structure **340** is disposed over the first conductive contact **290**. In some embodiments, the first interconnect structure **340** is physically and electrically connected to the first conductive contact **290**. The first interconnect structure **340** may include a first metal line **3402** and a first via **3404**; the first via **3404** is between the first conductive contact **290** and the first metal line **3402**. The first via **3404** connects the first metal line **3402** to the first conductive contact **290**. In some embodiments, the first interconnect structure **340** includes a plurality of first vias **3404** and a plurality of first metal lines **3402** stacked in an alternating manner to electrically connect the first conductive contact **290** to an external circuitry.

[0031] Each of the second conductive contacts **294** is disposed on the source region **260A** or the drain region **260B**. The first conductive contacts **290** and the second conductive contacts **294** may comprise same or different materials. For example, both the first conductive contacts **290** and the second conductive contacts **294** may include tungsten, copper, silver, titanium, polysilicon, or another suitable conductive material. In some embodiments, a silicide layer **302** is optionally

interposed between one of the second conductive contacts **294** and the source region **260A** and between another of the second conductive contacts **294** and the drain region **260B**. The silicide layer **302** acts as an ohmic contact layer to reduce a contact resistance between the source region **260A** or the drain region **260B** and the respective second conductive contact **294**.

[0032] The second interconnect structure **342** is disposed over the second conductive contacts **294**. The second interconnect structure **342** may include one or more second metal lines **3422** and one or more second vias **3424** connecting the second metal line **3422** to the second conductive contact **294**. In some embodiments, the first interconnect structure **340** and the second interconnect structure **342** may be simultaneously formed or formed using a same process.

[0033] The conductive plates **292** are disposed over the gate electrode **246**. In some embodiments, each conductive plate **292** is physically and electrically connected to the gate electrode **246** and extends toward the source region **260A** or the drain region **260B**. For example, each conductive plate **292** extends to an area between the gate electrode **246** and the source region **260A** or the drain region **260B** from the top-view perspective. The individual conductive plates **292** may have substantially same sizes and shapes. In some embodiments, electricity applied to the semiconductor device **10** from the first interconnect structure **340** is conducted to the conductive plates **292** through the first conductive contact **290** and the gate electrode **246**.

[0034] Referring to FIG. **1**, each conductive plate **292** has a first length **L1** in the X-direction and a second length **L2** in the Y-direction orthogonal to the X-direction. In some embodiments, the first length **L1** is greater than the second length **L2**. In some embodiments, the conductive plates **292** is evenly spaced in the Y-direction. The conductive plates **292** may be arranged with equal spacing between adjacent pairs along the Y-direction. In some embodiments, the conductive plates **292** are symmetric about a central axis **A** of the gate electrode **246**, wherein the central axis **A** extends in the Y-direction.

[0035] In some embodiments, the semiconductor device **10** further includes a first inter-layer dielectric (ILD) layer **270**, a second ILD layer **280**, an isolation layer **310**, and a low-k dielectric layer **320** disposed over the substrate **100**. The gate electrode **246** may be laterally surrounded by the first ILD layer **270**. The first conductive contact **290** and the conductive plates **292** are laterally surrounded by the second ILD layer **280**. The first interconnect structure **340** and the second interconnect structure **342** penetrate the isolation layer **310** and the low-k dielectric layer **320**. The first interconnect structure **340** in the isolation layer **310** and the low-k dielectric layer **320** may be formed using a dual-damascene method.

[0036] In some embodiments, the semiconductor device **10** further includes a first dielectric layer **210** and a second dielectric layer **220** disposed between the substrate **100** and the first ILD layer **270** and between the substrate **100** and the gate dielectric **232**. The substrate **100** and the first dielectric layer **210** may include a same material, such as silicon.

[0037] In some embodiments, the first dielectric layer **210** includes material provided by the substrate **100**. For example, the first dielectric layer **210** is a native oxide layer on the substrate **100**. The second dielectric layer **220** is deposited on the first dielectric layer **210**. Compared to first dielectric layer **210**, the second dielectric layer **220** may provide a higher thermal budget during fabrication of the semiconductor device **10**. The second conductive contacts **294** penetrate the first dielectric layer **210**, the second dielectric layer **220**, the first ILD layer **270**, and the second ILD layer **280**.

[0038] The semiconductor device **10** may further include spacer structures **250** and an etch stop layer **260**. In some embodiments, the spacer structures **250** are disposed on opposite sidewalls of the gate electric **232** and the gate electrode **246**. The etch stop layer **260** may be disposed between the second dielectric layer **220** and the first ILD layer **270** and between the spacer structures **250** and the first ILD layer **270**.

[0039] The semiconductor device **10** may further include a plurality of first isolation structures **110**, a plurality of second isolation structures **120**, a well region **150**, one or more guard regions **160**,

and a pair of lightly-doped regions **170** disposed in the substrate **100**. The first isolation structures **110** and the second isolation structures **120** may be shallow trench isolation (STI) structures. The first isolation structures **110** are used to define an active region **130** in and above which the semiconductor device **10** is formed. The second isolation structures **120** are disposed in the active region **130** and used to define a device region **140** in and above which the gate electrode **246**, the source region **260A**, and the drain region **260B** are formed.

[0040] The well region **150** is disposed in the active region **130** and may have a first conductivity type. The guard regions **160** may extend along two sides of the device region **140**. In some embodiments, each guard region **160** is disposed between one of the first isolation structures **110** and adjacent second isolation structure **120**. In alternate embodiments, the guard region **160** may form a ring encircling (from the top-view perspective of the substrate **100**) the device region **140** with the gate electrode **246**, the source region **260A** and the drain region **260B** therein. The guard regions **160** have the first conductivity type. In some embodiments, the guard regions **160** have an impurity concentration greater than that of the well region **150**.

[0041] The lightly-doped regions **170** are disposed in the device region **140** and spaced apart from each other in the X-direction. The lightly-doped regions **170** have a second conductivity type different from the first conductivity type. The lightly-doped regions **170** may be connected to the second isolation structures **120**, and the source region **260A** and the drain region **260B** are disposed at upper portions of the lightly-doped regions **170**. The source region **260A** and the drain region **260B** may have the second conductivity type. In some embodiments, in an n-type semiconductor device **10**, the first conductivity type is a p-type and the second conductivity type is an n-type. Alternatively, in a p-type semiconductor device **10**, the first conductivity type is the n-type and the second conductivity type is the p-type.

[0042] In some embodiments, the semiconductor device **10** is a metal-oxide semiconductor (MOS) transistor for medium voltage (MV) applications such as a source driver of a liquid-crystal display (LCD) monitor. The semiconductor device **10** may have an operating voltage of about 6 to 10 volts. Referring to FIGS. **1** to **3**, the distance **S** between the source region **260A** or the drain region **260B** and the gate electrode **246** is reduced by reducing a length, in the X-direction, of the gate electrode **246** while maintaining a desired channel length between the source region **260A** and the drain region **260B**.

[0043] With the reduction in the length of the gate electrode **246**, current caused by gate-induced drain leakage (GIDL) effect can be reduced during an off state of the semiconductor device **10**. However, a distribution of the highest electrical field is therefore moved from near the upper surface **102** of the substrate **100** to somewhere deeper in the semiconductor substrate **100**. Such a change in the distribution of the electrical field (along a channel region of the semiconductor device **10**) results in an increase of a substrate current during an on state of the semiconductor device **10**, which may cause power consumption performance to deteriorate during operation of the semiconductor device **10**.

[0044] To address the abovementioned issue, in the present disclosure, in order to rearrange distribution of the electrical field during the on state of the semiconductor device **10**, the conductive plates **292** are arranged in direct contact with the gate electrode **246** and extend to overlap some areas between the gate electrode **246** and the source region **260A** and between the gate electrode **246** and the drain region **260B**. For example, the distribution of the highest electrical field can be moved back to a position nearer to the upper surface **102** of the substrate **100** as compared to configurations in which the conductive plates **292** are absent. As a result, a substrate current during the on state is reduced.

[0045] In accordance with some embodiments of the present disclosure, FIG. **4** is a schematic top view of a semiconductor device **10A**, FIG. **5** is a schematic cross-sectional view along a line C-C' of the semiconductor device **10A** in FIG. **4**, and FIG. **6** is a schematic cross-sectional view along a line D-D' of the semiconductor device **10A** in FIG. **4**. Referring to FIGS. **4** to **6**, the semiconductor

device **10A** includes a substrate **100**, a gate dielectric **232**, a gate structure **245**, a source region **260A**, a drain region **260B**, a first ILD layer **270**, a second ILD layer **280**, a first conductive contact **290**, a plurality of second conductive contacts **294**, an isolation layer **310**, a low-k dielectric layer **320**, a first interconnect structure **340**, and a second interconnect structure **342**.

[0046] The gate dielectric **232** is disposed on an upper surface **102** of the substrate **100**, and the gate structure **245** is disposed over the gate dielectric **232**. The source region **260A** and the drain region **260B** are disposed in the substrate **100** and on opposite sides of the gate structure **245**. In some embodiments, the gate structure **245** includes a gate electrode **246** and a pair of conductive plates **293** disposed over the gate electrode **246**. The gate electrode **246** is disposed on the gate dielectric **232**. The gate dielectric **232** and the gate electrode **246** may be laterally surrounded by the first ILD layer **270**.

[0047] The conductive plates **293** are spaced apart from one another along the X-direction and connected to the gate electrode **246**. Each conductive plate **293** overlaps a portion of the gate electrode **246** and a portion of the first ILD layer **270**. In some embodiments, one of the conductive plates **293** overlaps an area AR1 between the gate electrode **246** and the source region **260A**, and another of the conductive plates **293** overlaps an area AR2 between the gate electrode **246** and the drain region **260B**. The conductive plates **293** are laterally surrounded by the second ILD layer **280**. In some embodiments, the gate electrode **246** has a first height H1, and the conductive plates **293** have a second height H2 greater than the first height H1.

[0048] Referring to FIG. 4, the substrate **100** includes an active area **130** defined by first isolation structures **110**. The active area **130** has a third length L3 in the Y-direction, and the conductive plate **292** has a fourth length L4 in the Y-direction. In some embodiments, the fourth length L4 is less than the third length L3. The gate electrode **246** may extend across the third length L3 of the active area **130**.

[0049] Referring to FIGS. 5 and 6, the gate electrode **246**, the first conductive contact **290**, and the first interconnect structure **340** are stacked along the Z-direction. In some embodiments, the first conductive contact **290** connects the first interconnect structure **340** to the gate electrode **246** and is laterally surrounded by the second ILD layer **280**. The first conductive contact **290** may have the second height H2. In some embodiments, the first conductive contact **290** and the conductive plates **293** are simultaneously formed or formed using a same process.

[0050] The second conductive contacts **294** and the second interconnect structure **342** may be stacked in the Z-direction. In some embodiments, each second conductive contact **294** connects the second interconnect structure **342** to the source region **260A** or to the drain region **260B**. The second conductive contacts **294** are electrically isolated from the first conductive contacts **290** and the conductive plates **293** by the first and second ILD layers **270** and **280**. The first and second interconnect structures **340** and **342** are surrounded by the isolation layer **310** and the low-k dielectric layer **320** disposed on the second ILD layer **280**.

[0051] FIG. 7 is a flowchart of a method **500** of manufacturing a semiconductor device **10**, in accordance with some embodiments of the present disclosure. FIGS. 8 to 33 are top views and cross-sectional views of intermediate stages of the method **500** of manufacturing the semiconductor device **10**, in accordance with some embodiments of the present disclosure. In the following description, the manufacturing stages shown in FIGS. 8 to 33 are discussed with reference to the process steps shown in FIG. 7. It should be understood that additional steps can be provided before, during, and after the steps shown in FIG. 7, and some of the steps described below can be replaced or eliminated, for additional embodiments of the method **500**. The order of the steps may be changed.

[0052] Referring to FIG. 8, a plurality of first isolation structures **110** and a plurality of second isolation structures **120** are formed in a substrate **100** in accordance with step S502 in FIG. 7. The substrate **100** has an upper surface **102** and a lower surface **104** opposite to the upper surface **102**. The first isolation structures **110** and the second isolation structures **120** extend from the upper

surface **102** for a first depth **D1** in the Z direction. The substrate **100** may be a bulk semiconductor substrate or a semiconductor-on-insulator (SOI) substrate. The substrate **100** may include crystalline silicon, crystalline germanium, a compound semiconductor (such as silicon carbide, gallium arsenide, or gallium phosphide), an alloy semiconductor (such as silicon germanium or gallium arsenide phosphide), or a combination thereof.

[0053] The first isolation structures **110** are formed in the substrate **100** to define various active regions **130**. The second isolation structures **120** may be located in the active region **130** and may be laterally offset from the first isolation structures **110** in the X-direction to define a device region **140**. In some embodiments, the active region **130** extends from a center portion of one of the first isolation structures **110** to a center portion of another of the first isolation structures **110**. The device region **140** may extend from a center portion of one of the second isolation structures **120** to a center portion of another of the second isolation structures **120**. The first isolation structures **110** and the second isolation structures **120** may be formed by etching the substrate **100** to form trenches and filling the trenches with an insulating material using a deposition operation. In some embodiments, a polishing operation is performed to remove excess portions of the insulating material from the upper surface **102** of the substrate **100**. The insulating material may include silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate (FSG), a low-k dielectric material, another suitable dielectric material, or a combination thereof.

[0054] In some embodiments, the first isolation structures **110** and the second isolation structures **120** have tapered profiles with their upper portions wider than their lower portions in a cross-sectional plane defined in the X-direction and Z-direction. Sizes of the first isolation structures **110** and the second isolation structures **120** may be substantially same or different. In some embodiments, the first isolation structures **110** have a first width **W1** (e.g., an upper or maximum width), and the second isolation structures **120** have a second width **W2** (e.g., an upper or maximum width) greater than the first width **W1**.

[0055] Referring to FIG. 9, a well region **150** is formed in the substrate **100** in accordance with step **S504** in FIG. 7. The well region **150** is formed in the active region **130**. In some embodiments, the well region **150** has a second depth **D2** greater than the first depth **D1** of the first and second isolation structures **110** and **120**. The well region **150** is formed by implanting an impurity of a first conductivity type into the substrate **100** through a first patterned mask layer **152**. After the well region **150** is formed, the first patterned mask layer **152** is removed, for example, in an ashing and/or wet strip operation.

[0056] Referring to FIG. 10, one or more guard regions **160** are formed in the substrate **100** in accordance with step **S506** in FIG. 7. The guard regions **160** may be disposed at an upper portion of the well region **150**. In some embodiments, the guard regions **160** are between the first isolation structure **110** and an adjacent second isolation structure **120**. The guard region **160** has a third depth **D3** less than the first depth **D1** of the first and second isolation structures **110** and **120**. In some embodiments, the well region **150** has a first doped concentration, and the guard regions **160** have a second doped concentration greater than the first doped concentration. The guard region **160** is formed by implanting ions of an impurity of the first conductivity type into the substrate **100** through a second patterned mask layer **162**. After the guard region **160** is formed, the second patterned mask layer **162** is removed, for example, in an ashing and/or wet strip operation.

[0057] Referring to FIG. 11, a plurality of lightly-doped regions **170** are formed in the substrate **100** in accordance with step **S508** in FIG. 7. The lightly-doped regions **170** are formed in the device region **140**. In some embodiments, the lightly-doped regions **170** are separated from one another by a first distance **S1** measured in the X-direction. The lightly-doped regions **170** have a fourth depth **D4**, which may be greater than the third depth **D3** and less than the first depth **D1**. The lightly-doped regions **170** are formed by implanting ions of an impurity of a second conductivity type into the substrate **100** through a third patterned mask layer **172**. The lightly-doped regions **170** may have a third doped concentration between the first and second doped concentrations. After the

lightly-doped regions **170** are formed, the third patterned mask layer **172** is removed, for example, in an ashing and/or wet strip operation.

[0058] Referring to FIG. **12**, a first dielectric layer **210** and a second dielectric layer **220** are subsequently formed on the substrate **100** in accordance with step S510 in FIG. **7**. The first dielectric layer **210** and the second dielectric layer **220** may be formed of oxide layers using different deposition methods. In some embodiments, the first dielectric layer **210** is formed using a thermal oxidation operation, and the second dielectric layer **220** is formed using a high temperature deposition, such as a rapid thermal chemical vapor deposition. Thus, the first dielectric layer **210** and the second dielectric layer **220** have different film properties such as different thermal budgets and different etching rates with respect to a same etchant. In some embodiments, the thermal oxidation operation is used to grow a high-purity and highly-conformal silicon oxide (SiO₂) such as silicon dioxide (SiO₂) on the substrate **100**, and the high temperature deposition is used to deposit silicon oxide with a high deposition rate on the first dielectric layer **210**, the first isolation structures **110**, and the second isolation structures **120**. In some embodiments, the first and second dielectric layers **210** and **220** have a combined thickness ranging from about 100 to about 300 angstroms. The first and second dielectric layers **210** and **220** may have a combined thickness of about 200 angstroms.

[0059] Still referring to FIG. **12**, a third dielectric layer **230**, a conductive layer **240**, and a fourth patterned mask layer **174** are subsequently formed on the second dielectric layer **220**. The third dielectric layer **230** may include a high-k material such as hafnium oxide (HfO₂) or other suitable material. The high-k dielectric material may have a dielectric constant greater than about 4. The third dielectric layer **230** is deposited to cover a substantially entire top surface of the second dielectric layer **220** using chemical vapor deposition (CVD) or other suitable methods.

[0060] The conductive layer **240** may be a polysilicon layer. The conductive layer **240** may be formed by any suitable technique, such as CVD, physical vapor deposition (PVD), atomic layer deposition (ALD), and/or other suitable methods. The fourth patterned mask layer **174** partially overlaps the well regions **150** and the lightly-doped regions **170**. In some embodiments, the fourth patterned mask layer **174** extends continuously from over one of the well regions **150** to over another of the well regions **150** (not shown).

[0061] Referring to FIGS. **13** and **14**, portions of the third dielectric layer **230** and the conductive layer **240** not protected by the fourth patterned mask layer **174** are removed. As a result, a gate structure **234** including a gate dielectric **232** and a dummy gate **242** is formed on the second dielectric layer **220** (step S512). The gate dielectric **232** and the dummy gate **242** extend in the Y-direction and across the active area **130**. The removal of the portions of the conductive layer **240** and the third dielectric layer **230** may be implemented using an etching operation. After the etching process, the fourth patterned mask layer **174** is removed, for example, using an ashing process or a wet etching process.

[0062] Referring to FIG. **15**, spacer structures **250** are formed on sidewalls of the gate dielectric **232** and the dummy gate **242** in accordance with step S514 in FIG. **7**. The formation of the spacer structures **250** may include blanket formation of a spacer layer (not shown) on the second dielectric layer **220**, the gate dielectric **232**, and the dummy gate **242**, and then performing one or more anisotropic etching operations to remove horizontal portions of the spacer layer. Remaining vertical portions of the spacer layer form the spacer structures **250**.

[0063] Referring to FIG. **16**, a source region **260A** and a drain region **260B** are formed in the substrate **100** in accordance with step S516 in FIG. **7**. The source region **260A** and the drain region **260B** are formed in the lightly-doped region **170**. The source region **260A** and the drain region **260B** are spaced apart from one another by a second distance S2 measured in the X-direction. In some embodiments, the second distance S2 is greater than a length L of the gate dielectric **232** or the dummy gate **242** in the X-direction.

[0064] An edge of the source region **260A** or the drain region **260B** is offset from an edge of the

dummy gate **242**. In some embodiments, the source region **260A** is separated from the gate dielectric **232** or the dummy gate **242** by a third distance **S3**. The drain region **260B** is also separated from the gate dielectric **232** or the gate electrode **242** by the third distance **S3**. In some embodiments, the third distance **S3** is in a range of about 40 nm to about 50 nm.

[0065] The source region **260A** and the drain region **260B** have a fifth depth **D5** that is less than the third depth **D3** of the guard region **160**. The source region **260A** and the drain region **260B** are formed by implanting ions of an impurity of the second conductivity type into the substrate **100** through a fifth patterned mask layer **176**. In some embodiments, the fifth patterned mask layer **176** covers the dummy gate **242** and the spacer structures **250**. The source region **260A** and the drain region **260B** have a fourth doped concentration greater than the first doped concentration of the well regions **150**.

[0066] In some embodiments, ions for the formation of the source and drain regions **260A** and **260B** are implanted with an implant energy between about 5 keV and about 8 keV, such as about 6 keV. The n-type source and drain regions **260A** and **260B** may be doped with, for example, phosphorus at a concentration of about 2×10^{15} atoms/cm². In some embodiments, ions for the formation of the source and drain regions **260A** and **260B** are implanted with an implant energy between about 2 keV and about 4 keV, such as about 3 keV when the second conductivity type is a p-type. The p-type source and drain regions **260A** and **260B** may be doped with, for example, boron at a concentration of about 6×10^{15} atoms/cm². After the source region **260A** and the drain region **260B** are formed, the fifth patterned mask layer **176** is removed, for example, in an ashing and/or wet strip operation.

[0067] Referring to FIG. 17, an etch stop layer **260** and a first ILD layer **270** are deposited over the second dielectric layer **220**, the dummy gate **242**, and the spacer structures **250** in accordance with step **S518** in FIG. 7. The etch stop layer **260**, having a substantially uniform thickness, has a topography following topographies of the second dielectric layer **220**, the dummy gate **242**, and the spacer structures **250**. In some embodiments, the etch stop layer **260** includes silicon nitride, silicon carbide (SiC), or other dielectric materials. The etch stop layer **260** may be formed by a suitable technique, such as CVD, PVD, ALD, other suitable methods, and/or a combination thereof.

[0068] The first ILD layer **270** is formed over the etch stop layer **260**. The first ILD layer **270** is blanketly formed to a height higher than a top surface of the dummy gate **242**. The first ILD layer **270** may include material such as tetraethylorthosilicate (TEOS), undoped silicate glass, phosphosilicate glass (PSG), boron doped silicon glass (BSG), borophosphosilicate glass (BPSG), fused silica glass (FSG), and/or other suitable dielectric materials. The first ILD layer **270** is formed by CVD, spin coating, sputtering, or other suitable methods.

[0069] Referring to FIGS. 18 and 19, a replacement gate operation is performed in accordance with step **S520** in FIG. 7. The replacement gate operation may include a planarization step, a removal step, and a deposition step. The planarization step is performed using, for example, a chemical mechanical polish (CMP) operation. The CMP operation is performed to remove excess portions of the first ILD layer **270** and the etch stop layer **260**, wherein the excess portions are over the dummy gate **242**. Accordingly, the dummy gate **242** is exposed.

[0070] In the removal step, the dummy gate **242** is removed to form a trench **244** with the spacer structures **250** as its sidewalls. The dummy gate **242** may be removed by a dry etch, a wet etch, or a combination thereof. In some embodiments, the dummy gate **242** is removed while the gate dielectric **232** is retained. In alternative embodiments, the dummy gate **242** and the gate dielectric **232** are removed during the removal step.

[0071] In the deposition step, the trench **244** is filled with one or more metal layers. The metal layers may include n-type work function metals (such as Ti, Ag, TaAl, TaAlC, TiAlN, TaC, TaCN, TaSiN, Mn, or Zr) and/or p-type work function metals (such as TiN, TaN, Ru, Mo, Al, WN, ZrSi₂, MoSi₂, TaSi₂, or NiSi₂). Other metal layers may also be formed in the trench **244**, such as a glue layer, a diffusion barrier layer, a filling layer, or the like. The work

function metal layer(s) or other metal layers may be deposited by CVD, PVD, electroplating and/or other suitable operations. After the deposition of the metal layers, a planarization process, e.g., CMP, is performed to planarize the metal layers to form a gate electrode **246** in the opening **244**. [0072] Referring to FIG. **20**, a second ILD layer **280** is deposited on the gate electrode **246** and first ILD layer **270** in accordance with step S522 in FIG. **7**. The first ILD layer **270** and the second ILD layer **280** may comprise same or different materials. The second ILD layer **280** may include oxide, material such as TEOS, undoped silicate glass, PSG, BSG, BPSG, FSG, and/or other suitable dielectric materials. The second ILD layer **280** is formed by CVD, spin coating, sputtering, or another suitable method. In some embodiments, the second ILD layer **280** may be planarized, using, for example, a CMP operation, to yield a flat surface.

[0073] Referring to FIGS. **21** to **23**, a first trench **284** and a plurality of second trenches **286** are formed through the second ILD layer **280** in accordance with step S524 in FIG. **7**. In some embodiments, the first trench **284** is disposed outside the active area **130** to expose a portion of the gate electrode **246**. The second trenches **286** are disposed inside the active area **130**. In some embodiments, each second trench **286** exposes a portion of the gate electrode **246**, a portion of the etch stop layer **260**, and a portion of the first ILD layer **270**. Referring to FIG. **23**, each second trench **286** may overlap an area of the lightly-doped region **170** between the gate electrode **246** and the source region **260A** or between the gate electrode **246** and the drain region **260B**. In some embodiments, the first trench **284** and the second trenches **286** are simultaneously formed in at least one etching operation. The second ILD layer **280** may be etched using a dry etch, a wet etch, or a reactive-ion etching (RIE) operation, for example.

[0074] Referring to FIGS. **24** to **26**, a first conductive material is deposited to fill the first trench **284** and the second trenches **286** in accordance with step S526 in FIG. **7**. Accordingly, a first conductive contact **290** and a plurality of conductive plates **292** are formed. In some embodiments, the first conductive material is conformally and uniformly deposited on the second ILD layer **280** and in the first and second trenches **284** and **286** until the first and second trenches **284** and **286** are filled. The first conductive material includes polysilicon or metal, such as tungsten, or the like. The first conductive material may be formed using CVD, PVD, ALD or another suitable operation. A planarization process is performed to remove portions of the first conductive material above an upper surface **2802** of the second ILD layer **280**. The excess conductive material may be removed from the second ILD layer **280** using, for example, a CMP operation.

[0075] Referring to FIGS. **27** and **28**, a pair of third trenches **288** are formed to expose the source region **260A** and the drain region **260B** in accordance with step S528 in FIG. **7**. In some embodiments, the third trenches **288** penetrate, from top to bottom, the second ILD layer **280**, the first ILD layer **270**, the etch stop layer **260**, the second dielectric layer **220**, and the first dielectric layer **210**. The third trenches **288** are formed using one or more etching operations.

[0076] Referring to FIG. **29**, a metal layer **300** is conformally deposited over the second ILD layer **280**, over the first conductive contact **290**, and in the third trenches **288**. The metal layer **300** may include titanium (Ti), cobalt (Co), nickel (Ni), tantalum (Ta), or platinum (Pt). In some embodiments, PVD is used for forming the metal layer **300**, although other methods, such as ALD, may also be used. In alternative embodiments, electroless plating, which can selectively form a metal layer on the source and drain regions **260A** and **260B**, but not on the second ILD layer **280** and the first conductive contact **290**, is used for forming the metal layer **300**.

[0077] Referring to FIG. **30**, an anneal operation is performed to react the metal layer **300** with silicon atoms in the silicon-containing substrate **100**, thus forming a silicide layer **302** at the source and drain regions **260A** and **260B** (step S530). Portions of the metal layer **300** that have not reacted with the silicon substrate **100** are removed through an etching method such as a wet etching.

[0078] Referring to FIG. **31**, a second conductive material is deposited to fill remaining portions of the third trenches **288** in accordance with step S532 in FIG. **7**. Accordingly, a pair of second conductive contacts **294** are formed. After the deposition of the second conductive material, a CMP

operation may be performed to planarize the upper surface **2802** of the second ILD layer **280**. In some embodiments, the first conductive material and the second conductive material may be same or different.

[0079] Referring to FIG. **32**, an isolation layer **310** and a low-k dielectric layer **320** are subsequently formed on the second ILD layer **280**, the first conductive contact **290**, the conductive plates **292**, and the second conductive contacts **294**. The isolation layer **310** may act as an etch stop layer. In addition, the isolation layer **310** may act as a silicidation blocking layer (SBL) to reduce formation of excess silicide during subsequent formation of conductive contacts, which may be useful in reducing current leaked through a path provided by the excess silicide, thereby improving device performance. In some embodiments, the isolation layer **310** includes TEOS. The isolation layer **310** may be formed using CVD, plasma-enhanced CVD, sputter, or other suitable methods.

[0080] The isolation layer **310** and the low-k dielectric layer **320** have different dielectric materials. In some embodiments, the low-k dielectric layer **320** includes, for example, PSG, BSG, BPSG, USG, FSG, spin-on-polymers, silicon carbon material, or a combination thereof. The low-k dielectric layer **320** may be deposited by spin coating, CVD, or another suitable operation. In some embodiments, a planarizing process can be optionally performed after the deposition of the low-k dielectric layer **320** to yield an acceptably flat topology.

[0081] Subsequently, a third trench **330** and one or more fourth trenches **332** are formed to penetrate the isolation layer **310** and the low-k dielectric layer **320**, wherein the first conductive contact **290** is exposed through the third trench **330** and one or more of the second conductive contacts are exposed through the fourth trenches **332**.

[0082] In accordance with step **S534**, a first interconnect structure **340** is formed to physically and electrically connect to the first conductive contact **290**, and a second interconnect structure **342** is formed to physically and electrically connect to the second conductive contact **294**, as illustrated in FIG. **33**. The first interconnect structure **340** and the second interconnect structure **342** are formed by depositing a metal material in the third trench **330** and the fourth trenches **332**. A planarization operation may be performed to remove excess portions of the metal material from a top surface of the low-k dielectric layer **320**. As a result, top surfaces of the first interconnect structure **340** and the second interconnect structure **342** are coplanar with the top surface of the low-k dielectric layer **320**. In some embodiments, the conductive plates **292** are electrically isolated from the first and second interconnect structures **340** and **342**.

[0083] In accordance with some embodiments of the present disclosure, a method of fabricating a semiconductor device is provided. The method includes steps of forming a source region and a drain region in a substrate; forming a gate structure on the substrate, wherein the source region and the drain region are disposed on opposite sides of the gate structure and separated from the gate structure by a distance; depositing an inter-layer dielectric (ILD) layer over the substrate and the gate structure; forming a first trench in the ILD layer, wherein the first trench exposes a first portion of the gate structure and overlies an area between the gate structure and one of the source and drain regions from a top-view perspective; and depositing a first conductive material in the first trench to form a conductive plate, wherein the conductive plate is electrically connected to the gate structure.

[0084] In accordance with some embodiments of the present disclosure, a method of fabricating a semiconductor device is provided. The method includes steps of forming a source region and a drain region in the substrate; forming a gate structure on a substrate, wherein the gate structure extends in a first direction and has a length in a second direction different from the first direction, the source region and the drain region are disposed on opposite sides of the gate structure, and a distance between the source region and the drain region in the second direction is greater than the length of the gate structure; depositing an ILD layer to cover the substrate and the gate structure; forming a first conductive contact penetrating the ILD layer and connected to the gate structure; and forming a plurality of conductive plates penetrating the ILD layer and connected to the gate

structure, wherein each conductive plate overlaps an area between the gate structure and one of the source and drain regions.

[0085] In accordance with some embodiments of the present disclosure, a semiconductor device is provided. The semiconductor device includes a substrate, a gate structure, a source region, a drain region, a pair of conductive plates, and a first conductive contact. The gate structure is disposed on a substrate and extends along a first direction, wherein the gate structure has a length in a second direction different from the first direction. The source region and the drain region are disposed in the substrate. The source and drain regions are separated from one another by a first distance greater than the length of the gate structure. The conductive plates are disposed over the gate structure, wherein each conductive plate extends in the second direction to overlap an area between the gate structure and the source region or the drain region. The first conductive contact is disposed on the gate structure. An electrical charge applied to the semiconductor device is conducted from the first conductive contact to the conductive plates through the gate structure.

[0086] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method of fabricating a semiconductor device, comprising: forming a source region and a drain region in a substrate; forming a gate structure on the substrate, wherein the source region and the drain region are disposed on opposite sides of the gate structure and separated from the gate structure by a distance; depositing an inter-layer dielectric (ILD) layer over the substrate and the gate structure; forming a first trench in the ILD layer, wherein the first trench exposes a first portion of the gate structure and overlies an area between the gate structure and one of the source and drain regions from a top-view perspective; and depositing a first conductive material in the first trench to form a conductive plate, wherein the conductive plate is electrically connected to the gate structure.
2. The method of claim 1, further comprising: forming a second trench penetrating the ILD layer to expose a second portion of the gate structure, wherein the first trench and the second trench are formed using a same process; and depositing the first conductive material in the second trench to form a first conductive contact.
3. The method of claim 2, further comprising: forming a plurality of first isolation structures in the substrate to define an active region, wherein the first conductive contact is outside the active region, and the conductive plate overlaps the active region from a top-view perspective.
4. The method of claim 3, further comprising: forming a pair of guard regions in the active region and on opposite sides of the source and drain regions, wherein the guard region has a first conductivity type and the source and drain regions has a second conductivity type different from the first conductivity type.
5. The method of claim 2, further comprising: forming a plurality of third trenches penetrating the ILD layer to expose the source region and the drain region; depositing a metal layer in the third trenches and onto the source region and the drain region; performing an annealing operation to react the metal layer with the source region and the drain region to form a silicide layer in the third trenches; and depositing a second conductive material in the third trenches over the silicide layer to form a second conductive contact.

6. The method of claim 5, further comprising, prior to the formation of the gate structure: forming a first dielectric layer on the substrate; and depositing a second dielectric layer on the first dielectric layer; wherein the third trenches penetrate the first and second dielectric layers, and the first dielectric layer comprises material provided by the substrate.
7. The method of claim 5, further comprising: forming a first interconnect structure over the ILD layer, wherein the first interconnect structure is physically and electrically coupled to the first conductive contact; and depositing a third dielectric layer on the ILD layer to laterally surround the first interconnect structure and cover the conductive plate.
8. The method of claim 1, wherein from a top-view perspective, the conductive plate extends from the gate structure toward the source or drain region to overlap an area between the gate structure and the source or drain region.
9. The method of claim 1, wherein the distance is in a range of about 40 nm to about 50 nm.
10. A method of fabricating a semiconductor device, comprising: forming a source region and a drain region in a substrate; forming a gate structure on the substrate, wherein the gate structure extends in a first direction and has a length in a second direction different from the first direction, the source region and the drain region are disposed on opposite sides of the gate structure, and a distance between the source region and the drain region in the second direction is greater than the length of the gate structure; depositing an ILD layer to cover the substrate and the gate structure; forming a first conductive contact penetrating the ILD layer and connected to the gate structure; and forming a plurality of conductive plates penetrating the ILD layer and connected to the gate structure, wherein each conductive plate overlaps an area between the gate structure and one of the source and drain regions.
11. The method of claim 10, wherein the first conductive contact and the conductive plates are simultaneously formed.
12. The method of claim 10, wherein the conductive plates are symmetric about a central axis of the gate structure, and the central axis is parallel to the first direction.
13. The method of claim 10, wherein the plurality of conductive plates extend toward one of the source region and the drain region and are evenly spaced in the first direction.
14. The method of claim 10, further comprising forming a plurality of second conductive contacts penetrating the ILD layer and connected to the source region and the drain region.
15. The method of claim 14, further comprising forming a first interconnect structure and a second interconnect structure over the ILD layer, the first conductive contact, the second conductive contacts, and the conductive plates, wherein the first interconnect structure is physically connected to the first conductive contact, and the second interconnect structure is physically connected to the second conductive contacts.
16. The method of claim 14, further comprising forming a metal silicide layer on the source region and the drain region prior to the formation of the second conductive contacts.
17. The method of claim 10, wherein a difference between the length of the gate structure and a distance between the source region and the drain region is in a range of about 80 nm to about 100 nm.
18. A semiconductor device, comprising: a gate structure disposed on a substrate and extending along a first direction, wherein the gate structure has a length in a second direction different from the first direction; a source region disposed in the substrate on a first side of the gate structure; a drain region disposed in the substrate on a second side of the gate structure opposite to the first side, wherein the source region are separated from the drain region by a first distance in the second direction and the first distance is greater than the length of the gate structure; and a pair of conductive plates disposed over and electrically connected to the gate structure, wherein each of the conductive plates extends in the second direction and overlaps an area between the gate structure and the source region or an area between the gate structure and the drain region.
19. The semiconductor device of claim 18, wherein the conductive plates are separated from each

other by a second distance less than the length of the gate structure.

20. The semiconductor device of claim 18, further comprising a first conductive contact disposed on the gate structure, wherein the first conductive contact and the conductive plates have a same height in a third direction different from the first and second directions.
